



GLOBAL HEALTH ANALYTICS

Global HIV Trends & Treatment Effectiveness

A data-driven assessment of incidence, mortality and treatment access across 150 countries

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Source: WHO Global Health Observatory | Tool: Microsoft Power BI | 2000–2024

What this presentation covers



1. Context & objectives

Why this analysis matters and the questions it answers



2. Data & method

The WHO dataset, the star-schema model, and the workflow



3. Data integrity

Limitations identified and handled transparently



4. Findings

The state of the epidemic and treatment effectiveness



5. Geographic view

Where treatment gaps and burden concentrate



6. Recommendations

Five evidence-led priorities for policymakers

Why this analysis matters



Three decades of investment

Prevention, awareness and antiretroviral therapy (ART) have turned HIV from a fatal disease into a manageable chronic condition for many.



Progress is uneven

Declining global mortality masks persistent regional disparities and fragile health systems where treatment access remains limited.



Effectiveness in question

Stakeholders lack a unified view to confirm whether interventions are working and where resources should be directed next.

Objectives & the questions we answer

Objectives



Determine whether new HIV infections are genuinely declining



Assess whether expanded ART access is reducing mortality



Identify regions and countries requiring urgent intervention

Key business questions

- How many new infections, deaths, and people living with HIV?
- What is treatment coverage, and the mortality & survival rate?
- How do infection and death trends compare over time?
- Which regions and countries carry the greatest burden?
- Does higher ART coverage correlate with lower mortality?
- Which countries are most at-risk from poor treatment access?

Data foundation & approach



Source & scope

WHO Global Health Observatory. 9,763 records · 150 countries · 6 WHO regions · 2000–2024.

Four indicators: new infections, HIV deaths, people living with HIV (PLHIV), and ART coverage.



Star-schema model

A clean dimensional model: one fact table related one-to-many to three dimensions: Geography, Indicator and Year.

Natural keys chosen deliberately: the dataset is small, clean and single-source.



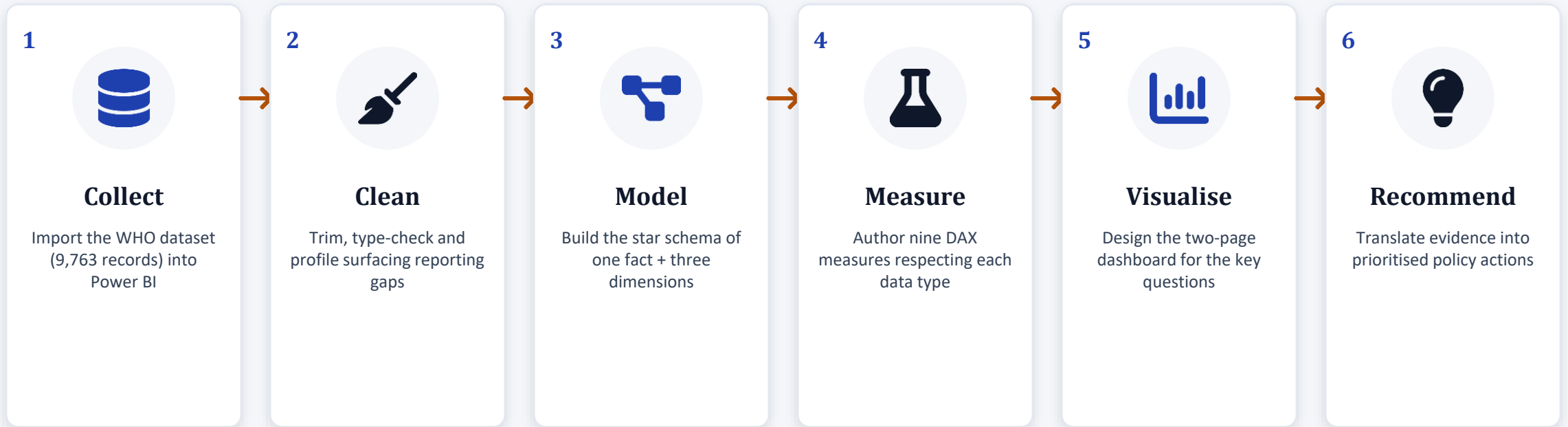
Measures & KPIs

Nine DAX measures, including a year-aware death rate, survival rate, and year-over-year growth each built to respect whether a figure is a count, a rate, or a stock.

WORKFLOW

From raw data to actionable insight

A repeatable, auditable pipeline — each stage feeds the next.



Limitations identified & handled transparently

Honest framing of what the data can and cannot support, disclosed up front, not hidden.



Irregular reporting frequency

Infections & ART coverage are reported every 5 years before 2020, then annually. Year-over-year growth is shown only for the annual window (2020–2024), where it is valid.



Censoring floor (value = 100)

~16% of records sit at a reporting floor of 100, a suppressed small count, not a precise measurement. Small-country figures are read as indicative, not exact.



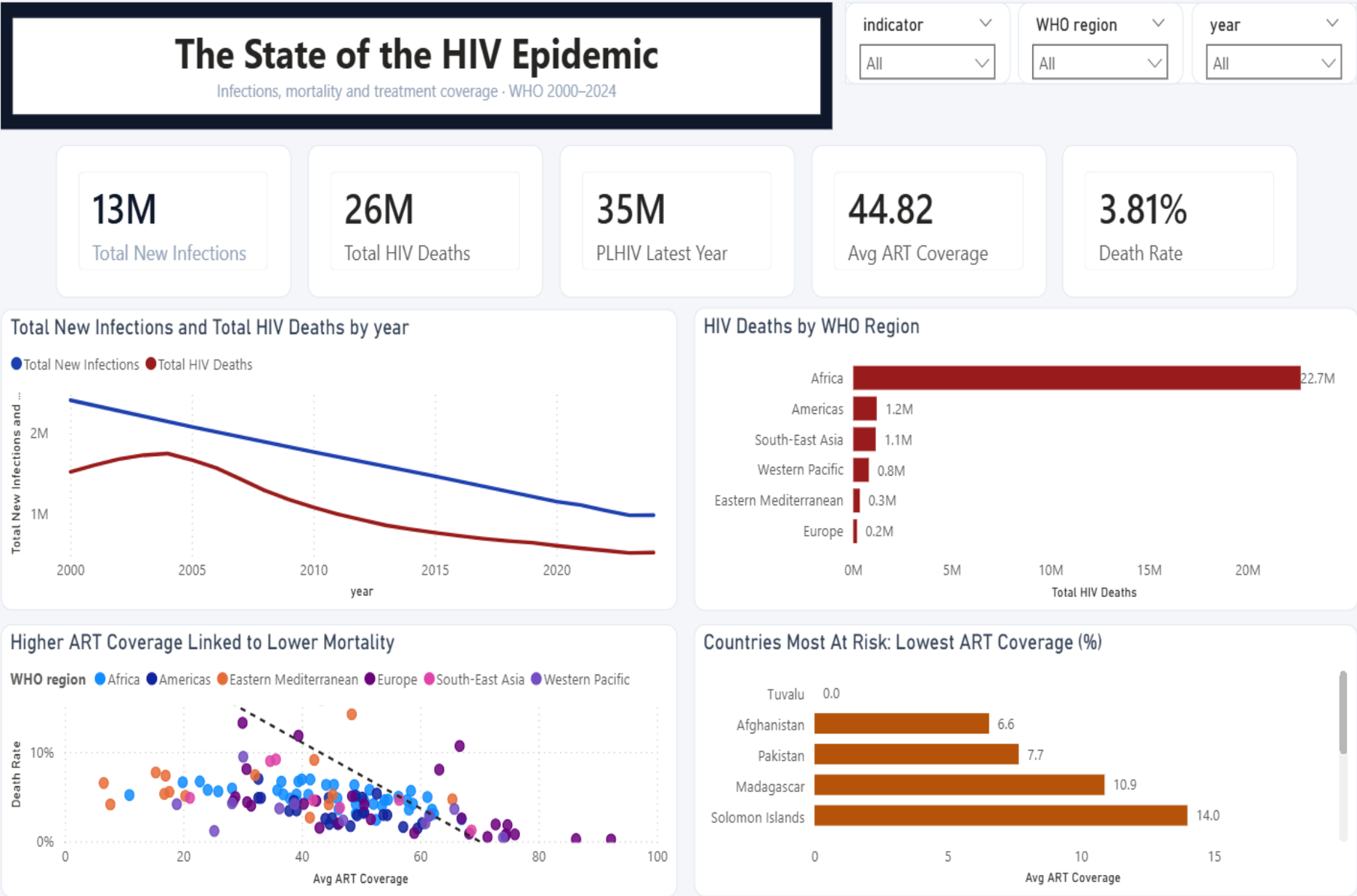
Stock vs flow distinction

PLHIV is a stock (a snapshot), so it is never summed across years; the latest-year value (~35M) is reported. Infections and deaths are flows and are summed legitimately.



Metric robustness

A deaths-per-1,000-infections ranking produced implausible values due to the reporting-frequency mismatch. It was replaced by an ART coverage ranking, which answers the 'poor access' question robustly.



The global HIV picture at a glance

13M

New infections

cumulative, reporting years

26M

HIV deaths

cumulative, 2000–2024

35M

People living with HIV

latest-year snapshot

44.82%

Avg. ART coverage

across countries

3.81%

Death rate

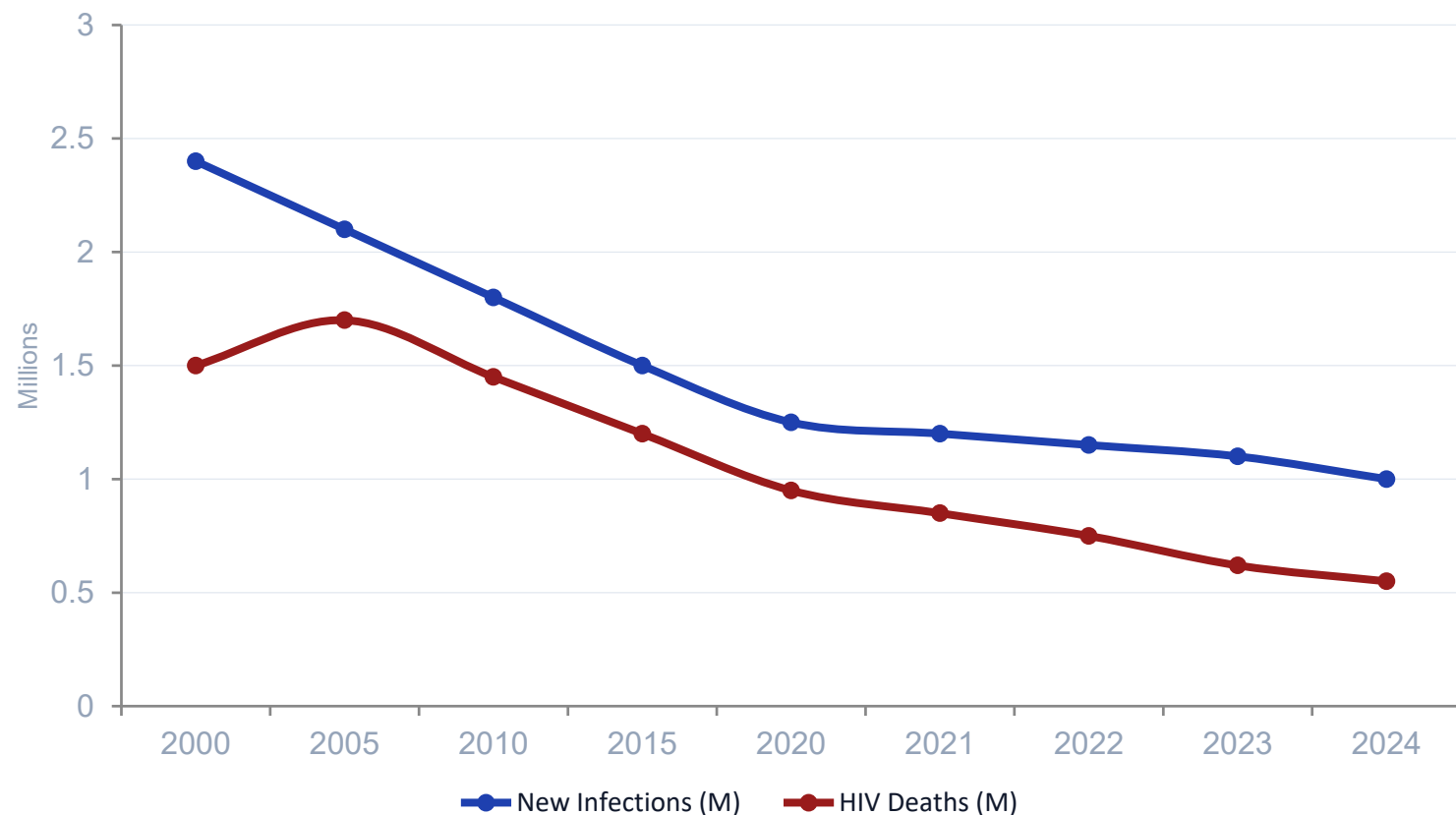
deaths ÷ prevalent PLHIV

96.19%

Survival rate

of people living with HIV

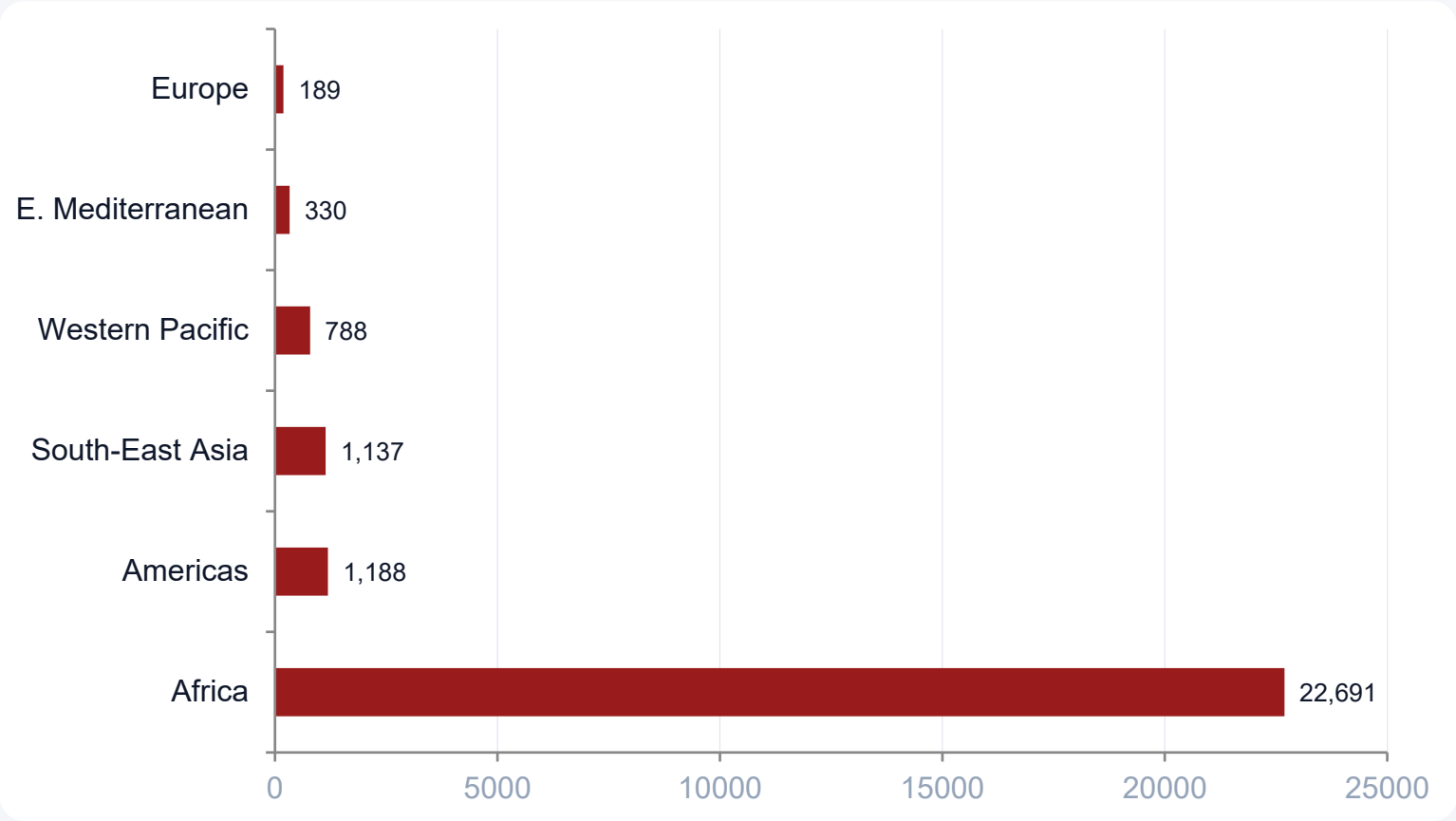
Are infections and deaths declining?



What it shows

- Both infections and deaths peaked in the mid-2000s and have declined since.
- In the reliable annual window, infections fell ~4–6% per year (2021–2023).
- Deaths fall faster than infections, the signature of treatment keeping people alive.

Where HIV mortality concentrates

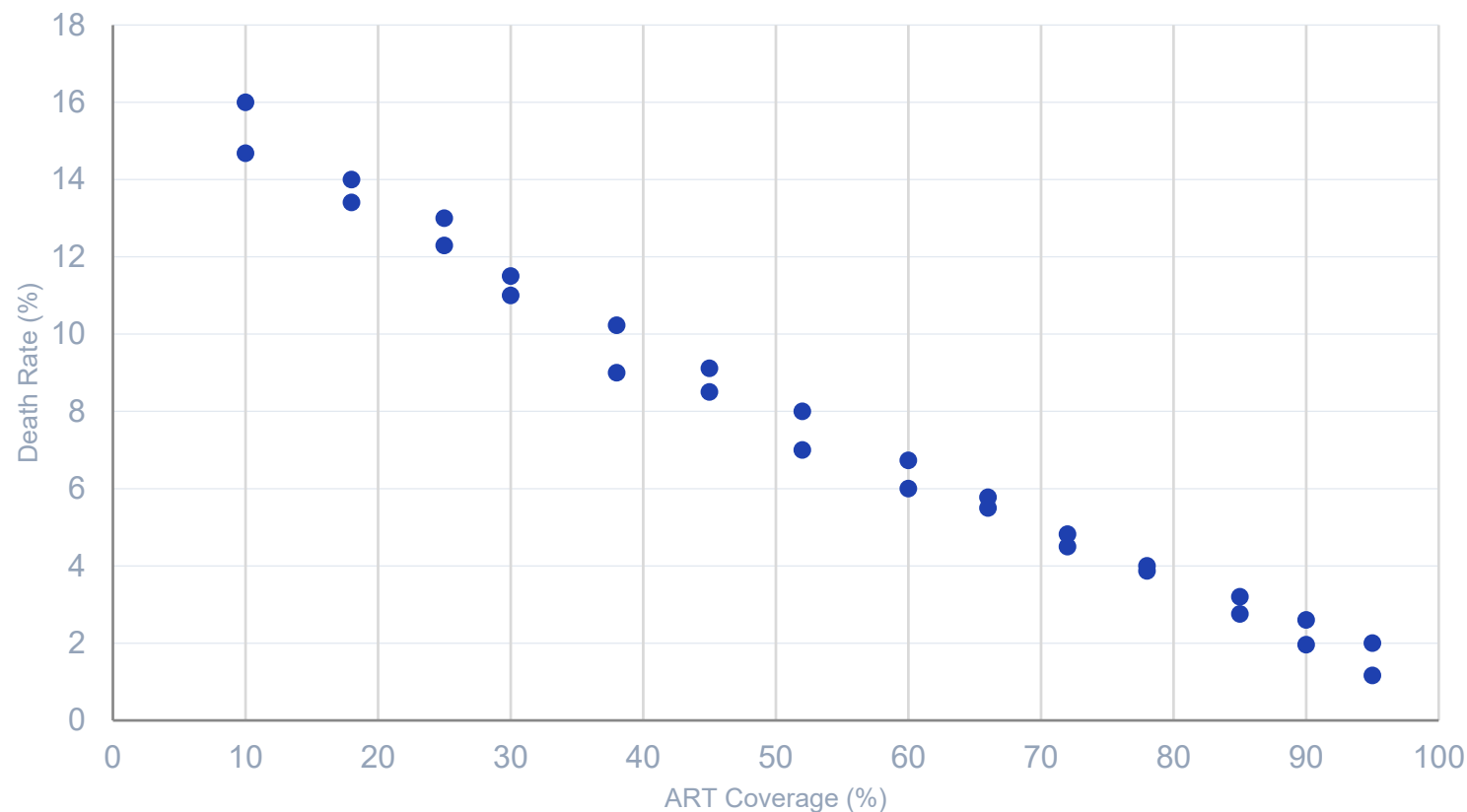


The headline

Africa carries ~22.7M cumulative deaths, roughly **19× the next region** (Americas, 1.2M).

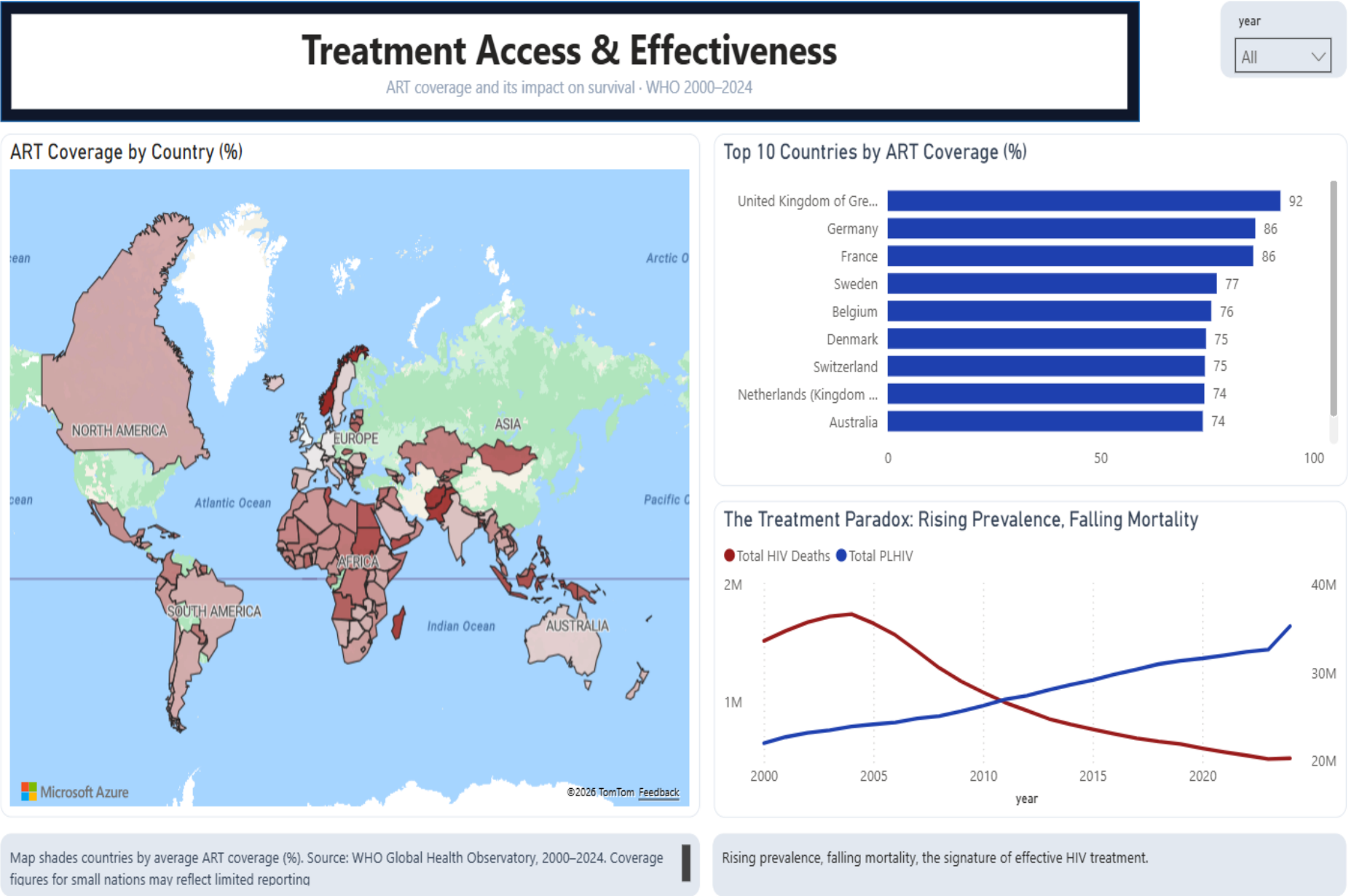
This concentration is the single clearest signal in the data for where resources must be directed.

Does higher ART coverage lower mortality?

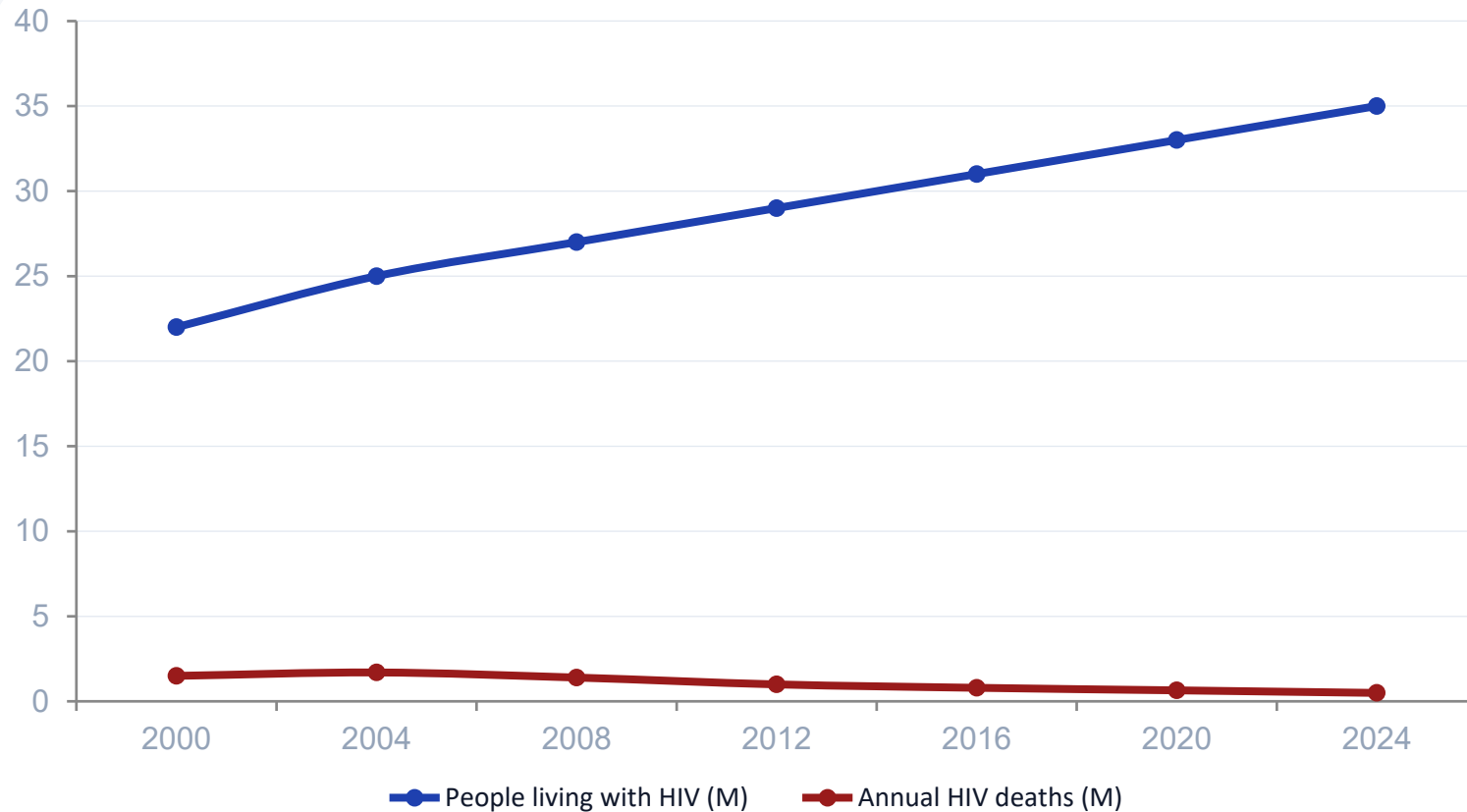


Interpretation

- A clear negative relationship: as ART coverage rises, the death rate falls.
- High-coverage countries cluster at low mortality (bottom-right).
- Read as ecological correlation, not causation; outliers flag access or quality gaps.



Rising prevalence, falling mortality



Why it matters

- More people are living with HIV every year, yet fewer are dying.
- Rising prevalence with falling mortality is the epidemiological signature of a disease shifting from fatal to chronic.
- 'More people living with HIV' is therefore a success indicator and a growing care obligation.

Four findings that define the picture



HIV is declining

Infections and deaths have fallen since their mid-2000s peak, though the pace of new-infection decline appears to be slowing in the most recent data.



Higher ART coverage, lower mortality

The relationship holds consistently across regions; a 96.19% survival rate reflects three decades of treatment scale-up.



Burden is deeply unequal

Africa carries ~19× the mortality of any other region — the dominant signal for where resources are needed.



Access gaps persist

Fragile and conflict-affected states still report near-zero ART coverage, leaving their populations acutely at risk.

Evidence-led priorities for policymakers



1. Target the funding core

Shift the global resource baseline to Sub-Saharan Africa, which bears ~22.7M cumulative deaths, far more than the rest of the world combined. Funding must track burden, not precedent.

Evidence: HIV Deaths by WHO Region



2. Deploy fragile-state delivery

Implement conflict-resilient, community-led ART networks in the lowest-coverage zones (Tuvalu, Afghanistan, Pakistan, Sudan), where standard clinical rollouts fail.

Evidence: At-Risk Countries · Coverage Map



3. Protect the treatment baseline

A 96.19% survival rate reflects sustained ART effectiveness, but new-infection decline appears to be slowing. Sustained ART procurement funding prevents a resurgence in mortality.

Evidence: Survival Rate KPI · YOY trend



4. Pivot to chronic-care systems

The Treatment Paradox shows a 35M cohort surviving longer. Transition infrastructure from acute epidemic response to sustainable, lifelong chronic-disease management.

Evidence: The Treatment Paradox



5. Systemic enabler: standardize data governance.

Move all indicators to annual reporting and reduce suppressed floor values, so future strategy rests on real-time, high-fidelity tracking.

Expected impact if recommendations are implemented



1. Higher ART coverage

More patients reached where the access gap is currently greatest



2. Lower mortality

Sustained ART scale-up compounds the mortality decline already under way



3. Better resource allocation

Funding follows verified burden by region, not historical precedent



4. Stronger surveillance

Annual reporting replaces irregular cycles, sharpening every decision



5. Improved reporting quality

Fewer suppressed floor values, higher confidence in country-level figures



From data to decisions

HIV is declining, and expanded treatment access is linked to lower mortality, but the burden remains profoundly unequal.

Targeted, evidence-led action in the highest-burden and lowest-access settings is the path forward.

Thank you

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Data source: WHO Global Health Observatory · #10Alytics · Health Tech Data Analyst Capstone